

Draft - expert review pending

This unit is complete and passes the platform's automated checks, but no reviewer has read it yet. Treat its sources and claims with care.

Draft translation - not yet human-reviewed / مسوده ترجمه

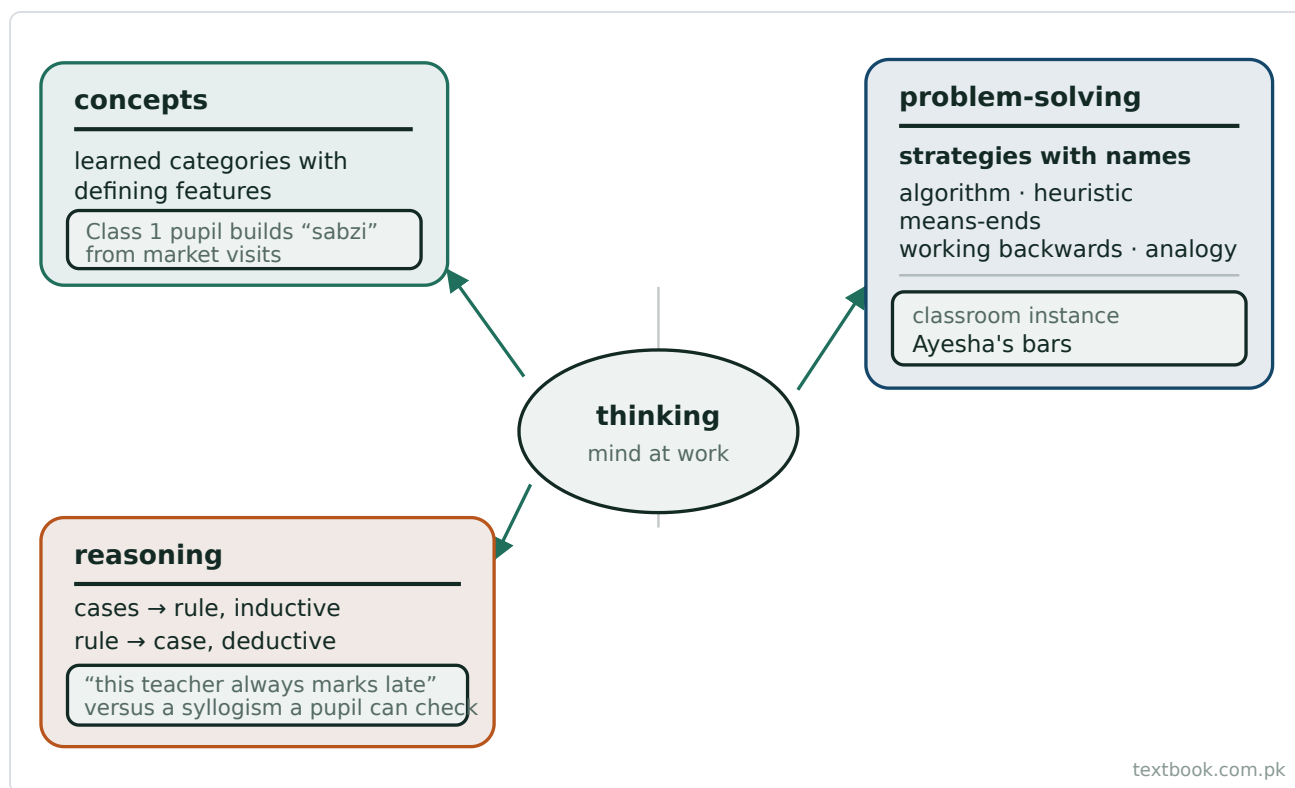
Topic 4.3 - Thinking, problem-solving and reasoning

A real classroom situation

Two Class 6 pupils meet the same word problem: "A rickshaw travels 24 kilometres in 40 minutes. At the same speed, how far in one hour?"

Ayesha reads it, draws the 40 minutes and the 24 kilometres as two bars, sees that 40 minutes is two-thirds of an hour, and answers 36. Bilal reads it, cannot see where to start, and writes $24 + 40 = 64$, "because you use the numbers".

Both pupils know the arithmetic. The difference is what their thinking *does* with a problem that does not announce its method. Ayesha has strategies; Bilal has a habit - use the numbers - that works often enough in a textbook to survive, and fails everywhere else.



textbook.com.pk

Explanation

Thinking and concept formation

Thinking is the manipulation of what the mind holds - and the units it manipulates are mostly **concepts**: learned categories that carry features and rules. "Sabzi", "fraction", "fair test", "polite request" - each is a concept, and each was built the same way: many encounters, the features gradually noticed, the boundaries gradually drawn.

Concept formation is where perception (Topic 4.1) meets memory (Topic 4.2). The Class 1 pupil building "sabzi" from market visits is forming a concept from examples; the pupil who then meets a carrot for the first time and files it correctly has a concept with a rule, not a list. Concepts are the archives of past learning and the tools of new thinking: the more precise a pupil's concepts, the less working memory each thought costs, because a category is one item instead of many.

For a teacher, two consequences. **Name the features.** A concept taught as a definition only ("a mammal is...") meets examples with no tools; a concept

taught through varied examples with the features named - and, just as usefully, near-misses (a whale is not a fish) - builds the rule. **Check the boundary.** The pupil who calls every four-legged animal a cat has a concept with a boundary problem, and the remedy is contrast, not repetition.

Problem-solving: strategies and obstacles

A problem, in the classroom sense, is a task whose method is not announced. Thinking that succeeds on such tasks runs on strategies that have names (Seifert & Sutton, 2009).

Algorithm: a guaranteed procedure - the long-division steps. Reliable, slow, and only available when one exists.

Heuristic: a rule of thumb - "find the question", "draw the situation", "try a simpler case". Fast, fallible, and the main content of "being good at problems".

Means-ends analysis: compare where you are with where you must be, do the step that shrinks the gap most, repeat. Ayesha's bars are means-ends made visible.

Working backwards: start from what is asked and ask what would produce it.

Analogy: this problem is like one I know - the strategy that makes transfer (Seifert & Sutton, 2009) the mark of understanding.

The obstacles have names too. **Functional fixedness:** seeing the object only in its usual role, so the scarf is never a fisherman's net. **Set effect:** the habit - Bilal's "use the numbers" - that solved yesterday's problems and blocks today's. And **working-memory overload:** a problem whose parts exceed the workbench is unsolved before it is read, which is why writing the givens down is not weakness but strategy.

The misconception to clear now. "Problem-solving is a gift." Every strategy above can be taught, named and practised; every obstacle above is a feature of everyone's thinking, including the teacher's. What separates Ayesha from Bilal is not a quantity of intelligence but a repertoire of moves and the habit of choosing among them - and repertoires are built.

Reasoning: inductive and deductive

Inductive reasoning runs from cases to a rule: every rickshaw fare I have checked ended in a zero, so they all do. It is the reasoning of generalisation, of science's early stages, and of the everyday classroom - and it is fallible by nature, because the next case can always break the rule.

Deductive reasoning runs from a rule to a case: all even numbers divide by two; this number is even; therefore it divides by two. It is the reasoning of proof, and its classroom value is not the mathematics but the check: a deduction that fails tells you the rule is wrong, not that the world is misbehaving.

Pupils use both daily and need both named. The everyday generalisation ("this teacher always marks late") is induction, and its remedy is a counter-example, not an argument. The syllogism a pupil can check ("if the test is Friday and today is Thursday, then...") is deduction, and its remedy is showing the step. Confusing the two is the commonest reasoning error in a staff room: treating a generalisation as a proof, or a proof as an insult.

Problem-solving strategies

strategy	an everyday instance	one obstacle it can hit
Algorithm	the long-division steps	no algorithm exists for this problem, so the pupil is stuck
Heuristic draw it, try a simpler case	Ayesha's bars for the rickshaw problem	the drawing becomes the task and the question is forgotten
Means-ends analysis	compare where you are with the answer, shrink the biggest gap	the gap chosen is the easy one, not the blocking one
Working backwards	start from what is asked: what would produce it?	the start point itself is unclear, so the backwards chain has no first link
Analogy	this is like the water-pump question from last term	the likeness is superficial and the method is imported wrongly

textbook.com.pk

textbook.com.pk

Class activities that work with attention and memory

The guide's Week 9 slot is class activities, and the whole unit now supplies the design rules. An activity that works **with** the mind: one attention-heavy demand at a time (Topic 4.1); a load that fits the workbench, with the parts written down (Topic 4.2); strategies named aloud so they become concepts (this topic); and retrieval practised at a spacing the timetable can hold (Topic 4.2).

The redesigned lesson looks like this: the problem arrives with the question named and the givens listed (attention aimed, memory offloaded); the class tries one strategy aloud and names it ("we are working backwards"); near-misses are welcomed as boundary work on concepts; and the week's plan returns to the problem once more, days later, as a two-minute retrieval opener. Nothing in that design costs money. All of it costs planning - which is the unit's argument about where ability actually comes from.

Activity: Name the move

In pairs, about 20 minutes.

Solve each of the following mentally, then name the strategy you used, and one obstacle that could have blocked you.

1. A shopkeeper sells three pens for Rs 45. What would seven pens cost?
2. A class of 40 needs one lab bench for every 5 pupils. The lab has 6 benches. What must change?
3. A bus leaves Karachi at 8:00 and reaches Hyderabad at 10:30. The distance is about 160 kilometres. Roughly what speed does it average?
4. You have a 3-litre jug and a 5-litre jug. How do you measure exactly 4 litres?

Then compare your strategies with your partner's. Where you used different moves on the same problem, decide which move the problem was *built* to teach.

Check your understanding

1. Define a concept in your own words, and explain the boundary problem with the "billi" example. (*Understand*)

2. Name any three problem-solving strategies with an instance of each.
(Remember)
3. Bilal writes $24 + 40 = 64$. Name the obstacle and the strategy that would have helped, and say how you would teach it. (Apply)
4. Distinguish inductive from deductive reasoning with one classroom example of each. (Understand)
5. Design, in three lines, a class activity that works with attention and memory rather than against them, naming the rule each line follows. (Apply)

Summary

Thinking runs on concepts - learned categories whose features can be taught through examples and near-misses - and on named strategies for problems whose method is not announced: algorithm, heuristic, means-ends, working backwards, analogy. Its obstacles - functional fixedness, set effect, overload - are features of everyone's thinking, and they answer to teaching, not to talent. Reasoning runs in two directions, from cases to rules (inductive) and rules to cases (deductive), and each has its own remedy when it fails. Class activities that work with the mind aim the beam, fit the workbench, name the strategy, and space the retrieval.

Self-assessment checklist

Sign in to sync your ticks across devices.

- ☐ I can define a concept and teach one through examples and near-misses.
- ☐ I can name five problem-solving strategies and one obstacle for each.
- ☐ I can distinguish inductive from deductive reasoning with a classroom example of each.
- ☐ I can design one activity that follows the unit's design rules and name the rule each part follows.

Try this at your practicum school

Teach one lesson in which you name the strategy aloud every time the class uses one ("we are drawing it", "we are working backwards"). At the end, ask the class to list the moves they made today. Keep the list: it is the beginning of a strategy wall, and the pupils' own words are the concepts.

Summative task

Write **250-300 words** on Bilal's 64.

Analyse Bilal's answer as a thinking problem rather than an ability verdict: name the obstacle, the missing strategies, and what his habit tells you about the problems he has been set. Then design the next lesson's first ten minutes - the activity, the strategy named, the load on the workbench - and justify each choice from this unit.

Mini-rubric

Criterion	What a strong answer shows
Analysis, not verdict	The answer is read through strategies and obstacles, not "he is weak".
Design discipline	Each element of the ten minutes is justified by a named rule from the unit. (<i>Analyze</i>)
Habit explained	The answer says what the textbook's same-shaped problems taught him.
Plain, accurate writing	Terms used correctly; no padding; within the word count.

Further reading

- Seifert, K., & Sutton, R. (2009). *Educational Psychology* (2nd ed.), Chapter 9. University of Manitoba. CC BY.

<https://open.umn.edu/opentextbooks/textbooks/educational-psychology>

- Vosniadou, S. (2001). *How children learn*. Educational Practices Series No. 7. International Academy of Education / UNESCO IBE. <https://eric.ed.gov/?id=ED455955>

Tip: select any text on this page first to comment on that specific passage - or leave nothing selected for general feedback.

Give feedback